

# Cauchy–Kovalevskaya, Riquier, and the parametric data of ansatz 5

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A companion note to *A New Method of Solving PDEs*, expanding the remark that “Riquier’s existence theorem gives a unique analytic solution for any consistent choice of the parametric (free) data — the rigorous Cauchy–Kovalevskaya generalization.” It records what the principal/parametric split actually looks like — the staircase picture — and works it out concretely for the hydrogen PDE under ansatz 5, where the parametric data collapses to thirteen constants and Riquier’s theorem degenerates to ODE existence.

## 1 The Cauchy–Kovalevskaya theorem

The Cauchy–Kovalevskaya (CK) theorem is the fundamental existence-and-uniqueness theorem for analytic PDEs. Consider a system in *CK form* — solved for the highest pure derivative in one distinguished variable, say  $t$ :

$$\frac{\partial^k u}{\partial t^k} = F(t, x, u, \dots)$$

where  $F$  is analytic in its arguments and the right-hand side involves only derivatives of order  $\leq k$  having fewer than  $k$  derivatives in  $t$ . If one prescribes *analytic Cauchy data* — the  $k$  functions  $u, \partial_t u, \dots, \partial_t^{k-1} u$  on the hypersurface  $t = 0$ , as analytic functions of  $x$  — then a *unique analytic solution* exists in a neighborhood of the hypersurface.

The proof is formal power series plus convergence: the equation determines every Taylor coefficient of  $u$  from the Cauchy data (each application of  $\partial_t^k$  is given by lower data), and the method of majorants shows the series converges. Cauchy proved the first-order case (1842); Kovalevskaya gave the general formulation in her 1875 dissertation [1].

Two limitations matter here:

- **Only the “solved” (determined) form.** CK says nothing about overdetermined systems — several PDEs on one unknown, where cross-differentiation produces integrability conditions.
- **Analytic category only.** Lewy’s example [5] shows existence can fail outright in the smooth category, and Kovalevskaya’s heat-equation example shows the series can diverge when the equation is not in CK form.

## 2 Riquier’s generalization

Riquier’s theorem [2] extends CK to *orthonomic (passive) systems*. Once an overdetermined system has been completed to passive form — all integrability conditions accounted for, which is exactly

what Rosenfeld–Gröbner or Thomas decomposition produces — the derivatives split, relative to the ranking, into:

- **principal** derivatives: those in the cone generated by the leaders (derivatives of the leading derivatives), determined by the equations; and
- **parametric** derivatives: everything else.

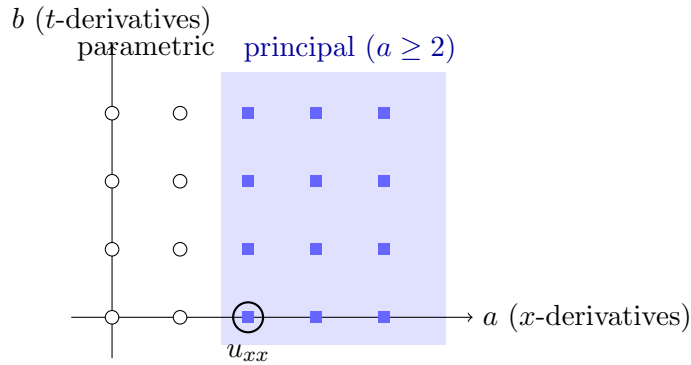
Riquier: prescribe the parametric derivatives' values (analytic initial data respecting the ranking) freely; there is then a unique analytic solution. The parametric derivatives of a regular differential chain genuinely parameterize its solution set, and counting them (the differential dimension / Hilbert–Kolchin polynomial [6]) measures the size of the solution space.

### 3 The staircase picture — when is the parametric set finite?

A natural first reaction is that CK data (a set of *functions*) is far more data than Riquier's parametric data (a set of *parameters*). It is not: the two carry the same amount of data, and the parametric set is finite only in a special case.

Identify each derivative  $\partial_x^a \partial_t^b u$  with the lattice point  $(a, b) \in \mathbb{N}^2$ . A leader at  $(a_0, b_0)$  claims as principal the shifted quadrant  $(a_0, b_0) + \mathbb{N}^2$ . The parametric derivatives are the uncovered lattice points. The complement of a union of shifted quadrants is finite *iff the leaders include a pure power of every variable* — a leader on each coordinate axis (the monomial-ideal fact:  $\mathbb{N}^n \setminus I$  is finite iff  $I$  contains some  $x_i^{k_i}$  for every  $i$ ).

**Heat equation.** Take  $u_{xx} = u_t$ , ranked so the leader is  $u_{xx}$ , at  $(2, 0)$ :



Nothing caps the  $t$ -axis, so the parametric set is the two infinite columns  $a \in \{0, 1\}$ . Prescribing all Taylor coefficients in column  $a=0$  is prescribing the arbitrary analytic function  $u(0, t)$ ; column  $a=1$  is  $u_x(0, t)$ . One infinite strip  $\leftrightarrow$  one free *function*. This is exactly CK data for the equation read with  $x$  as the evolution direction — on a determined system, Riquier specializes to CK, same data re-indexed.

**Overdetermined case.** Now a system whose leaders are  $u_{xx}$  and  $u_{tt}$  (integrability conditions already satisfied):



- **each constant:** only the order-0 value survives. Eleven parameters — but not free ones (§5).

Where the raw Schrödinger equation alone (one leader in 3 variables) leaves two infinite slabs — Cauchy data of two arbitrary analytic functions of two variables, dimension polynomial growing like  $s^2$  — ansatz 5 grinds the staircase down until the Hilbert–Kolchin polynomial is eventually constant. Total parametric data:  $\Psi(p_0)$ ,  $\Psi'(p_0)$ , and the 11 constants — thirteen numbers, modulo the two homogenization scalings (`homogenize_groups = (Avars, Vvars)` in `helium.sage`), which are gauge.

## 5 Where Riquier’s theorem lands on ansatz 5

Riquier’s “unique analytic solution for any *consistent* choice of the parametric data” plays out in two layers on ansatz 5:

1. **Consistency is where all the work lives.** The chain-rule equations are automatically integrable (cross-derivatives agree because  $v$  is explicit and  $v_{xy} = v_{yx}$ ). The genuine condition comes from reducing the Schrödinger PDE modulo the ansatz by Ritt reduction [4] (`differential_prem` in `joca.sage`): the remainder must vanish identically, and its coefficients — as a polynomial in  $\Psi$ ,  $\Psi'$  and the coordinates — form a system of *algebraic* equations in the 11 constants. That polynomial system is exactly the input whose minimal primes `minprimes.sage` computes. Consistency of the parametric data = the constants lie on that variety.
2. **Existence then costs nothing.** For each point of each prime component, Riquier hands a unique analytic solution per choice of the two remaining truly-free parameters  $\Psi(p_0)$ ,  $\Psi'(p_0)$  — which is just the classical fact that a second-order linear ODE has a 2-dimensional solution space. On this ansatz, Riquier degenerates all the way to ODE existence; no genuine PDE-analytic content remains.

That is the shape of the whole program in staircase terms: the ansatz converts “solve a PDE” (infinite-dimensional parametric data, degree-2 dimension polynomial) into “find the variety where finite parametric data is consistent” (commutative algebra in 11 variables). Every hydrogen solution the ansatz can see corresponds to a point on one of the minimal primes plus two ODE integration constants.

## References

- [1] S. von Kowalevsky, *Zur Theorie der partiellen Differentialgleichungen*, J. Reine Angew. Math. **80** (1875), 1–32.
- [2] C. Riquier, *Les systèmes d’équations aux dérivées partielles*, Gauthier-Villars, Paris, 1910.
- [3] M. Janet, *Leçons sur les systèmes d’équations aux dérivées partielles*, Gauthier-Villars, Paris, 1929.
- [4] J. F. Ritt, *Differential Algebra*, AMS Colloquium Publications **33**, American Mathematical Society, 1950.
- [5] H. Lewy, *An example of a smooth linear partial differential equation without solution*, Ann. of Math. **66** (1957), 155–158.

[6] E. R. Kolchin, *Differential Algebra and Algebraic Groups*, Academic Press, New York, 1973.

/Claude (Fable 5)